

1. Key Deliverables

- **High-Level Design (HLD):** Outlines overall architecture, modules, database design, and technologies.
- **Low-Level Design (LLD):** Detailed breakdown including logic flows, class diagrams, and algorithms.
- **Data Models:** Entity Relationship Diagrams (ERDs) or schema definitions for data storage.
- **API Specifications:** Documentation for system communication (endpoints, request methods, formats).

2. Tools & Techniques

- **UML:** Visual modeling for system architecture (Class/Sequence/Activity diagrams).
- **Figma:** Wireframing, prototyping, and UI/UX visual layout creation.
- **Lucidchart/Draw.io:** Creating flowcharts, network topologies, and data flow diagrams.

3. System Design vs. UI/UX Design

Feature	System Design	UI/UX Design
Focus	Architecture, backend, scalability, security.	User journey, aesthetics, accessibility.
Goal	Robustness, performance, maintainability.	Intuitive, usable, pleasing experience.
Outputs	Architecture diagrams, API specs, schemas.	Wireframes, prototypes, style guides.

4. Poor Design Example: Healthcare.gov (2013)

- **The Issue:** A rigid "hub-and-spoke" architectural design created single points of failure, failing to handle massive concurrent traffic during launch.
- **Outcome:** The system crashed immediately, requiring a massive "tech surge" to rewrite backend architecture, costing significant taxpayer funds and causing reputational damage.

Sources

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